How a Finding Is Computed

A plain-language walkthrough — no formulas

The one rule that matters most: the AI never calculates anything about the machine's sensors. Every number you see — a vibration level, an anomaly score, a time range — was computed by ordinary, checkable Python code before the AI ever sees it. The AI's only job is to turn those already-computed results into a sentence you can read, and to say exactly where each one came from.

Step by step, in plain language

1. The recording is checked.

Before anything else happens, the file is checked for the right columns, a valid header, and readable numbers (see the quarantine guide). Only files that pass are used.

2. The signal is cut into short windows.

Each sensor channel is sliced into windows of 2,048 consecutive readings, each overlapping the last by half. Think of it like reading a long strip of paper through a small moving window instead of trying to take in the whole strip at once. (Every channel is recorded as a raw voltage, not a pre-converted physical unit — see the Machine Fact Sheet — so “strong” or “unusual” in a Finding means unusual for that channel's own voltage range, not a calibrated physical quantity like g-force or newtons.)

3. Each window gets a set of measurements.

For every window, the system computes standard signal statistics — how strong the signal is, how spiky it is, how it's distributed — and, when the timing data allows it, which frequency ranges carry the most energy. See the glossary for what each one means.

4. "Normal" is defined separately for every condition.

Because spindle speed, feed and depth change the machine's behaviour on purpose, the system never compares a reading to one single global average. Instead, it builds a separate "what's normal here" profile for every combination of condition and channel, using only the recordings that passed the quality check.

5. A window is flagged if it's far from what's normal for its condition.

The system measures how far each window's measurements are from its own condition's normal profile. A window that's unusually far away becomes a **Finding**. Nothing is flagged just because it's different from a different condition's numbers — that comparison would be meaningless.

6. Every Finding remembers exactly where it came from.

A Finding always carries its recording file, which channel, and the exact range of samples (and, when available, the exact time range) that produced it. That's what a citation like [Finding 1] points to — you can always trace it back to the original data.

7. The AI explains, it doesn't invent.

When you ask a question, the system hands the AI only the Findings that are relevant, already phrased in plain language, with their citations attached. The AI writes a short, readable explanation and marks each claim with the Finding it came from. It is never given the raw sensor numbers to interpret on its own, and it is never asked to do arithmetic.

What this means when you read an answer

Every specific claim you see — a vibration reading being unusual, a value moving in a certain direction — should have a small citation chip like [Finding 1] next to it. Clicking that chip shows you exactly which recording, which channel and which stretch of time it's based on. If a sentence has no citation attached to it, treat it as general explanation rather than a specific measured claim.

Source: README.md §2 ("How it works, end to end"), rewritten at a simpler reading level. Model version referenced in citations: mahalanobis-diagonal-v3.